# ENHANCING MEDICAL LLMS WITH FINE-TUNING, KNOWLEDGE DISTILLATION AND RETRIEVAL-AUGMENTED GENERATION

**Muhammad Awais Naeem, Xinyuan Huang, Wei-Chun Chang, Mohamad Chehade**
{awais.naeem, hxydiana, weichun.chang, chehade}@utexas.edu

## ABSTRACT

Parameter Efficient Fine-tuning (PEFT) has shown to turn large language models (LLMs) into domain experts, while Knowledge Distillation (KD) transfers knowledge from larger or better performing models to smaller alternatives, offering a scalable path toward efficient and specialized AI. On the other hand, Retrieval-Augmented Generation (RAG) utilizes external memory to further enhance a model's domain expertise. In this paper, we propose combining these techniques to improve models' performance in the medical domain by: (1) distilling knowledge into worse-performing models and fine-tuning them with ground truth, and (2) introducing a retrieval-augmented KD framework that utilizes bidirectional alignment between teacher and student embeddings. Experiments reveal that combining PEFT, KD and RAG improves student model performance in the medical domain. Our findings suggest that KD and RAG can be powerful enhancement to PEFT for building scalable, high-performing, and domain-adapted LLMs for medicine discourse.

## 1 INTRODUCTION

LLMs like `GPT`, `Llama`, and `Grok` each have strengths in different domains—`GPT` excels at math, while `Grok` is better suited for social media analysis. Understanding these differences is especially crucial in high-stakes fields like medicine, where accuracy and safety are paramount. Moreover, as these models have different strengths, each model can benefit from various fine-tuning techniques Hu et al. (2021); Dettmers et al. (2023); Li & Liang (2021), distillation Xu et al. (2024) and retrieval-augmented generation Lewis et al. (2021) using the responses from other language models.

This study is designed to address the following key research questions:

- How do popular LLMs perform in medical domain in a zero-shot setting?
- How does vanilla PEFT and knowledge distillation techniques enhance models' performance in the medical domain?
- Can models effectively combine the knowledge learned during fine-tuning on domain-specific datasets and the distilled knowledge from other language models?
- Can models use other models' generated responses as external memory to conduct retrieval augmented generation?

Through these questions, we aim to explore how combining PEFT, KD and RAG could bring additional performance enhancement, particularly in the medical domain, and provide insights into the efficiency of knowledge transfer between models.

### 1.1 BENCHMARKING OPEN-SOURCE LLMS ON MEDICAL DATASETS

To get a clear picture of how well state-of-the-art LLMs perform in the medical domain, we start by evaluating them on two healthcare datasets without any prompt engineering or fine-tuning. By feeding questions into the different models and comparing their responses to the ground-truth answers, we get an unfiltered look at their out-of-the-box capabilities on these medical question answering

tasks of different flavors. And this baseline serves as the foundation for the second phase of our study.

### 1.2 Fine-tuning with various approaches

In the second phase of our study, we aim to explore the fine-tuning capabilities of various models, specifically GPT-4o (OpenAI (2024)) and LLaMA 3.2 (AI (2024)). The fine-tuning process will be conducted using two medical question-answer datasets to assess the performance and adaptability of these models in a specialized domain. We plan to fine-tune using both ground truths and generated responses from a well-performing model, effectively achieving knowledge distillation.

### 1.3 Retrieval-augmented knowledge distillation

In the third phase, we leverage generated answers from one model as external context to support another model in producing its own answer. This setup allows us to explore how sharing generated knowledge across models can enhance performance and offers insights into effectively combining the "wisdom" of multiple models.

## 2 Related work

**Fine-tuning techniques.** Instead of full-parameter fine-tuning, parameter-efficient tuning has proven to achieve computational efficiency. Common techniques include: Adapters Houlsby et al. (2019), Low-Rank Adaptation (LoRA) Hu et al. (2021) and QLoRA Dettmers et al. (2023). LoRA enabled effective adaptation by injecting trainable low-rank matrices into each layer while keeping the original weights frozen. This approach significantly reduces the number of trainable parameters and memory requirements, making it suitable for resource-constrained scenarios. Building on this, QLoRA extends LoRA by combining it with 4-bit quantization of the base model weights, allowing full fine-tuning of large models on a single GPU without compromising performance. In this paper, we adapt LoRA and QLoRA to achieve efficient fine-tuning given compute constraints.

**Retrieval-augmented Generation.** Retrieval-Augmented Generation (RAG) is a technique that combines retrieval-based methods with generative models to enhance the quality and accuracy of generated outputs. Instead of relying solely on pre-trained knowledge, a RAG model first retrieves relevant documents or passages from an external knowledge source based on the input query. These retrieved pieces of information are then provided as additional context to the generative model, which uses them to produce more informed and grounded responses. In our paper, answers generated from a model is used as the external knowledge source and provided as context to another model.

**Medical LLMs.** Several studies have explored different fine-tuning approaches for medical LLMs. LlamaCare Li et al. (2024) , a fine-tuned version of Llama 2 for the clinical domain, was trained using the MIMIC-III dataset and GPT-4-generated instructions. This study highlights the effectiveness of instruction fine-tuning in adapting LLMs to complex medical scenarios. On the other hand, OpenMedLM Maharjan et al. (2024) explored robust prompting techniques as an alternative to fine-tuning for optimizing open-source LLMs on medical benchmarks like MedQA and PubMedQA.

## 3 Methodology

Our fine-tuning methodology consists of three main tasks:

**Task 1: Distillation of Knowledge to Another Model**

The process begins by extracting questions from the medical datasets, followed by parsing them and generating answers using baseline models such as GPT-4o and LLaMA 3.2. The generated responses are then compared against the ground truth answers provided in the dataset. Based on this comparison, the under-performing model is identified to be fine-tuned using the responses from the superior model. For instance, if GPT-4o outperforms LLaMA 3.2, the answers generated from the GPT-4o model serve as training data for further fine-tuning of LLaMA 3.2.

This process allows us to examine whether the newly introduced information enhances the performance of LLaMA 3.2. After fine-tuning, the updated LLaMA 3.2 model is compared with its orig-

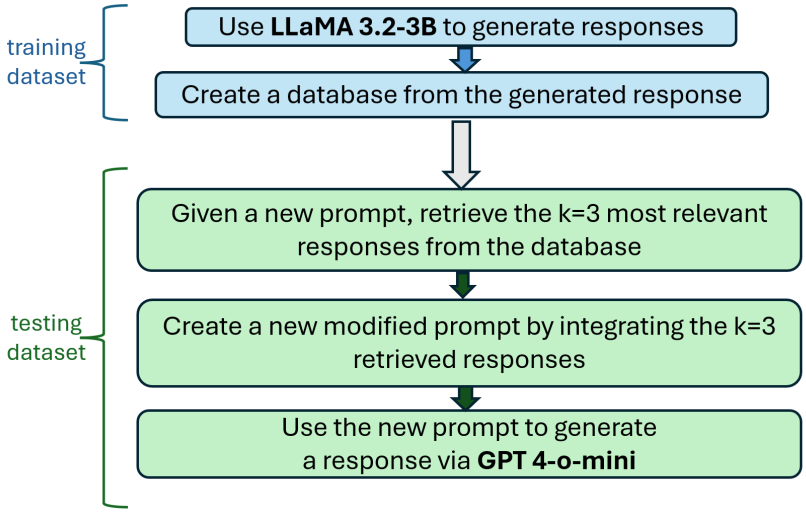



Figure 1: An example of enriching the prompts passed to GPT 4-o-mini with responses generated from LLaMA 3.2-3B.

inal version and GPT-4o to assess whether knowledge has been effectively transferred or encoded within the model.

**Task 2: Evaluating Model Learning from Fine-tuning**

This task involves fine-tuning the language models (LLaMA3.2, GPT-4o) using the ground truth dataset, and evaluating the performance to determine whether fine-tuning has resulted in an improvement, demonstrating that the model is learning effectively from the fine-tuning process. A slightly different approach will be taken here to analyze if the knowledge distilled in a model from another model can be combined with the knowledge enhanced during fine-tuning on the relevant dataset.

To accomplish this, we will take the the distilled LLaMA3.2 (or GPT-4o) model fine-tuned in *Task 1* and further fine-tune the ground-truth dataset rather than taking the base model. The superior model in *Task 1* will still be a base model and fine-tuned on the ground truth dataset. A comparable performance of both fine-tuned models will mean that the distilled model effectively encoded the information from another model plus the ground-truth dataset.

**Task 3: Creating a `RAG` system from synthetic answers** In this task, we employ zero-shot inference in one model to generate responses for the training data set and then collect these in a vector database by chunking them into multiple sentences. These chunked synthetic answers then act as the knowledge base of RAG system for *other model* during the evaluation on test dataset. For each question in the test dataset, we retrieve the top $k$ semantically similar chunks related to the original query from the knowledge base and then add them to the prompt as an additional context. For example, we apply LLaMA 3.2-3B to the *training* dataet to create a database of chunked synthetic responses. Then, we use this database in creating *enriched* prompts for the testing dataset, for which GPT 4-o-mini will provide responses. This methodology is further illustrated in Figure 1.

By performing this task, we are examining how much the *zero-shot* performance of a model improves by simply adding relevant context (to the query) in the prompt with the context essentially coming from the zero-shot inference of *another model*. We also explore whether this technique can match the performance achieved through knowledge distillation and, more importantly, fine-tuning with the ground truth data (Task 2).

# 4 EXPERIMENTS

We employed two small-to-medium-sized models in our experiments: llama-3.2-3B and gpt-4o-mini. These two models are chosen based on our constraints on compute and the model availability.

## 4.1 DATASETS

Two medical datasets are used to evaluate the vanilla LLMs and their enhanced versions. The "doctor-patient dialogue" dataset Kabatubare (2024) tests the coherence and relevance of the response in addition to domain knowledge, while The "medical-exam-style short answer questions" FreedomIntelligence (2024) focuses more on domain knowledge and have ground truth answers with lower variability. Given the open-ended nature of both datasets, BLEU Papineni et al. (2002) and ROUGE Lin (2004) scores will be used.

## 4.2 ESTABLISHING THE BASELINE

To establish the baseline performance of the two models unchanged, we perform zero-shot inference on two datasets. We performed inference on the full datasets but in different comparison scenarios only the test split results are used to calculate the metrics.

## 4.3 FINE-TUNING WITH GROUND-TRUTH DATA

We fine-tune each model on each dataset seperately using only the training split of the datasets. For GPT-4-o-mini, we use the OpenAI fine-tuning functions on the API platform OpenAI (2025). For LLaMA-3.2-3B, we use Unsloth Unsloth-Team (2024) library for parameter efficient fine-tuning (LoRA) Hu et al. (2021) to fit within the compute resources available.

## 4.4 KNOWLEDGE DISTILLATION

This phase of our project involves distilling the knowledge of stronger model into the weaker model. Specifically, we use answers generated by the better-performing model and then use those answers to fine-tune the weaker-performing model. This is to test the model's ability to learn from a stronger model. Unlike phase 1 which used the entire dataset for evaluation, we split the dataset into train and test based on 90:10 ratio. The dataset is collected from phase 1 evaluation (i.e. original set of questions + generated responses by stronger model). We evaluate the distilled model only on the test set.

## 4.5 RETRIEVAL AUGMENTED GENERATION (RAG)

We perform this task on the two datasets separately, and consider two versions: (i) Using the GPT 4-o-mini chunked responses to create the knowledge base and then using the retrieved chunks as additional context while prompting LLaMA 3.2-3B on test data questions, and (ii) Using LLaMA 3.2-3B responses to create the vector database and then prompting GPT-4o-mini by including additional context for questions in test dataset. A template for the enriched prompt is shown in Figure 2.

To form the knowledgebase for RAG approach, we use an open-source vector database Chroma Contributors (2025). To divide the responses into individual chunks, we use *RecursiveTextSplitter* from LangChain Harrison Chase (2022), another open-source framework to integrate LLMs into different applications. To form the embeddings of the individual chunks, we employ the default sentence transformer installed with the Chroma i.e., *all-MiniLM-L6-v2* Reimers & Gurevych (2019). Chunk size is decided by looking at the average character length of all the answers in each individual dataset to ensure that each answer is averagely divided into separate chunks.

# 5 RESULTS

**Zero-shot performance:** Performance evaluation of GPT-4o-mini and Llama-3.2 on the two datasets is presented in Table 1. The results show that GPT–4o-mini baseline (titled "GPT-Base"

```
You are a concise, highly knowledgeable medical
assistant. Provide medically accurate and
direct answers to health-related questions
without any disclaimers or explanations.

Context:
{context}

Question: {question}

Answer:
```

Figure 2: A template for the new prompt after adding past responses obtained via retrieval.

Table 1: BLEU and ROUGE metrics on the "Kabatubare" and "FreedomIntelligence" datasets.

| Dataset | Model | BLEU | ROUGE-1 | ROUGE-2 | ROUGE-L | ROUGE-LSum |
|---|---|---|---|---|---|---|
| Kabatubare | LLaMA-Base | 0.0086 | 0.1731 | 0.0333 | 0.0928 | 0.1260 |
| | GPT-Base | 0.0129 | 0.2255 | 0.0427 | 0.1174 | 0.1540 |
| FreedomIntelligence | LLaMA-Base | 0.0074 | 0.1138 | 0.0394 | 0.1082 | 0.1092 |
| | GPT-Base | 0.0093 | 0.1155 | 0.0411 | 0.1043 | 0.1049 |

in the table) outperforms Llama-3.2 baseline (titled "LLaMA-Base" in the table) on Dataset 1 while achieving on-par performance on Dataset 2. As a result, we distill GPT-4o-mini into Llama-3.2 3B.

**Knowledge Distillation:** Distillation using Dataset 1 ("LLaMA-Base-Distilled") shows that Llama model can successfully learn from GPT generated responses and achieve similar performance as GPT. Note that the metrics for the base models in Table 2 are recalculated from only the test set data, to ensure a valid comparison.

**Fine-tuning on ground-truth data:** For the first two datasets, Table 3 shows that fine-tuning with actual data outperforms distillation, as expected. However, what is interesting is that the distilled LLaMA model, when fine-tuned, is able to match, or even exceed, the performance of the GPT model.

**Using retrieval for enriching prompts:** Table 3 shows that the retrieval approach is able to **greatly improve the zero-shot performance** of a model using another model for Dataset 1 ("Kabatubare"). Furthermore, for Dataset 2, this approach is able to match the fine-tuned models. Such results suggest a parameter-free, computational-friendly mechanism to improve the performance of benchmark models on medical datasets.

## 6 Discussion

The results highlight the effectiveness of combining PEFT, KD, and RAG to enhance LLM performance on medical QA tasks. Distilling knowledge from GPT-4o-mini into LLaMA 3.2-3B significantly reduced the performance gap, as evidenced by BLEU and ROUGE scores. Further fine-tuning of the distilled LLaMA model on ground-truth data allowed it to match or exceed the fine-tuned GPT model, indicating that distillation can serve as an effective pre-training strategy.

Retrieval-based prompt enrichment (RAG) also proved valuable. In zero-shot scenarios, it enabled LLaMA to approximate the performance of fine-tuned GPT, particularly on the FreedomIntelligence dataset. This suggests that RAG offers a lightweight, compute-efficient alternative for performance gains. However, fine-tuning remained superior on tasks demanding contextual coherence, such as those in the Kabatubare dataset.

Table 2: BLEU and ROUGE metrics on the "Kabatubare" dataset.

| Dataset | Model | BLEU | ROUGE-1 | ROUGE-2 | ROUGE-L | ROUGE-LSum |
|---|---|---|---|---|---|---|
| Kabatubare | LLaMA-Base | 0.0084 | 0.1711 | 0.0327 | 0.0909 | 0.1249 |
| | GPT-Base | 0.0129 | 0.2239 | 0.0422 | 0.1163 | 0.1527 |
| | LLaMA-Base-Distilled | 0.0121 | 0.2221 | 0.0403 | 0.1165 | 0.1515 |

Table 3: BLEU and ROUGE metrics on the "Kabatubare" and "FreedomIntelligence" datasets: comparing fine-tuning with prompt enrichment via retrieval.

| Dataset | Model | BLEU | ROUGE-1 | ROUGE-2 | ROUGE-L | ROUGE-LSum |
|---|---|---|---|---|---|---|
| Kabatubare | LLaMA-Base | 0.0085 | 0.1712 | 0.0328 | 0.0919 | 0.1249 |
| | GPT-Base | 0.0129 | 0.2239 | 0.0421 | 0.1163 | 0.1527 |
| | LLaMA Fine-tuned | 0.0200 | 0.2147 | 0.0523 | 0.1429 | 0.1427 |
| | GPT Fine-tuned | 0.0244 | 0.2066 | 0.0339 | 0.1199 | 0.1200 |
| | LLaMA Distilled Fine-tuned | 0.0200 | 0.2147 | 0.0523 | 0.1429 | 0.1427 |
| | GPT Enriched | 0.0152 | 0.2343 | 0.0403 | 0.1267 | 0.1450 |
| | LLaMA Enriched | 0.0148 | 0.2196 | 0.0394 | 0.1272 | 0.1363 |
| FreedomIntelligence | GPT-Base | 0.0096 | 0.1125 | 0.0411 | 0.1055 | 0.1058 |
| | LLaMA-Base | 0.0107 | 0.1329 | 0.0477 | 0.1290 | 0.1290 |
| | GPT Fine-tuned | 0.0847 | 0.2955 | 0.1267 | 0.2898 | 0.2899 |
| | LLaMA Fine-tuned | 0.1027 | 0.2439 | 0.1000 | 0.2398 | 0.2390 |
| | GPT Enriched | 0.0978 | 0.2888 | 0.1258 | 0.2874 | 0.2875 |
| | LLaMA Enriched | 0.0919 | 0.2458 | 0.0972 | 0.2427 | 0.2429 |

Limitations include reliance on automated metrics, which may not fully reflect clinical correctness or safety, and heuristic chunking methods that could be improved with adaptive retrieval strategies. Synthetic responses used in RAG may also risk propagating hallucinations or model bias.

Future directions include exploring bidirectional distillation with mutual reasoning and developing integrated RAG-KD frameworks. Broader dataset coverage and expert evaluations will be essential for assessing generalizability and clinical reliability.

## 7 Conclusion

We have explored the effect of fine-tuning benchmark language models on medical question-answering tasks. Specifically, we considered GPT 4-o-mini and LLaMA 3.2-3B and applied them to 2 common medical datasets. Our projected is divided into 4 tasks: zero-shot performance evaluation, knowledge distillation from the stronger model to the weaker model, fine-tuning with ground truth data, and using an external retrieval-based memory built from the synthetic responses from one model and using it to enrich the prompts passed to the new model. The results show that combining these 3 techniques improves the performance of both models in the medical domain. They also show that retrieval can be a cheap alternative to the computationally expensive parameter fine-tuning.

## 8 Team Contribution

**Xinyuan Huang:** Ideation, Documentation, GPT fine-tuning and inference on Kabatubare Dataset

**Mohamad Chehade:** Documentation, Illustrations, GPT fine-tuning and inference on the Freedom Intelligence Dataset

**Wei-Chun Chang:** Documentation, LLaMA fine-tuning and inference on the Freedom Intelligence Dataset

**Muhammad Awais Naeem:** Ideation, Documentation Review, LLaMA fine-Tuning and inference on Kabatubare/Freedom Intelligence Dataset

## REFERENCES


Meta Meta AI. Llama 3: Open-weight large language model. https://ai.meta.com/llama, 2024. Accessed: 2025-03-10.

Chroma Contributors. Chroma: Open-source vector database for ai applications. https://www.trychroma.com, 2025. Accessed: 2025-04-29.

Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. Qlora: Efficient finetuning of quantized llms. *Advances in neural information processing systems*, 36:10088–10115, 2023.

FreedomIntelligence. Medical dataset 2. https://huggingface.co/datasets/FreedomIntelligence/medical-o1-verifiable-problem, 2024. Accessed: 2025-03-10.

Harrison Chase. Langchain. https://github.com/langchain-ai/langchain, 2022.

Neil Houlsby, Andrei Giurgiu, Stanislaw Jastrzebski, Bruna Morrone, Quentin de Laroussilhe, Andrea Gesmundo, Mona Attariyan, and Sylvain Gelly. Parameter-efficient transfer learning for nlp. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, pp. 2790–2799, 2019. URL http://proceedings.mlr.press/v97/houlsby19a.html.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685*, 2021.

Kabatubare. Medical dataset 1. https://huggingface.co/datasets/Kabatubare/medical, 2024. Accessed: 2025-03-10.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive nlp tasks, 2021. URL https://arxiv.org/abs/2005.11401.

Rumeng Li, Xun Wang, and Hong Yu. Llamacare: An instruction fine-tuned large language model for clinical nlp. In *Proceedings of the LREC 2024 Conference*, 2024. URL https://aclanthology.org/2024.lrec-main.930.pdf.

Xiang Lisa Li and Percy Liang. Prefix-tuning: Optimizing continuous prompts for generation, 2021. URL https://arxiv.org/abs/2101.00190.

Chin-Yew Lin. ROUGE: A package for automatic evaluation of summaries. In *Text Summarization Branches Out*, pp. 74–81, Barcelona, Spain, July 2004. Association for Computational Linguistics. URL https://aclanthology.org/W04-1013/.

Jitendra Maharjan, Ananya Garikipati, N. P. Singh, et al. Openmedlm: Prompt engineering can out-perform fine-tuning in medical question-answering with open-source large language models. *Scientific Reports*, 14:14156, 2024. doi: 10.1038/s41598-024-64827-6. URL https://doi.org/10.1038/s41598-024-64827-6.

OpenAI. Gpt-4o: Openai's multimodal model. https://openai.com/research/gpt-4o, 2024. Accessed: 2025-03-10.

OpenAI. Fine-tuning, 2025. URL https://platform.openai.com/docs/guides/fine-tuning. Accessed: 2025-04-29.

Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. Bleu: a method for automatic evaluation of machine translation. In Pierre Isabelle, Eugene Charniak, and Dekang Lin (eds.), *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics*, pp. 311–318, Philadelphia, Pennsylvania, USA, July 2002. Association for Computational Linguistics. doi: 10.3115/1073083.1073135. URL https://aclanthology.org/P02-1040/.

Nils Reimers and Iryna Gurevych. Sentence-bert: Sentence embeddings using siamese bert-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 11 2019. URL https://arxiv.org/abs/1908.10084.

Unsloth-Team. Unsloth: Efficient fine-tuning of large language models. https://github.com/unslothai/unsloth, 2024. Accessed: 2025-03-10.

Xiaohan Xu, Ming Li, Chongyang Tao, Tao Shen, Reynold Cheng, Jinyang Li, Can Xu, Dacheng Tao, and Tianyi Zhou. A survey on knowledge distillation of large language models, 2024. URL https://arxiv.org/abs/2402.13116.